

- Q.29 Explain types and applications of Stainless steel
- Q.30 Name few properties and application of Pig iron.
- Q.31 Write few uses of engineering material in chemical industry.
- Q.32 Explain ferrous metals in brief with examples.
- Q.33 Define and classify Ceramics material.
- Q.34 Explain in brief varieties of Iron ores
- Q.35 Explain in brief properties of
- Clay
  - Stonewares

#### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain in brief about rubber and polycarbonates.
- Q.37 Explain importance of engineering materials and Bureau of Indian standard (BIS) for engineering materials.
- Q.38 Write a short note on any two:
- Types of uses of Copper alloy.
  - HDPE and LDPE
  - Types of Glass

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**3rd Sem / Chem**

**Subject:- Introduction to Engineering Materials / Engg. Materials**

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Corrosion is
- Decay of metal due to reaction with environment
  - both (a) and (c)
  - accumulation of metal
  - None of above
- Q.2 Name of aluminium ore is:
- Tin
  - Limonite
  - Bauxite
  - None of above
- Q.3 Brittleness is
- Lack of ductility
  - both a & b
  - excess ductility
  - none of the above
- Q.4 Copper is used for making electrical wires due to its good:
- Ductility
  - Brittleness
  - Malleability
  - None of above

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- Q.5 Pig iron is produced from furnace known as:  
a) Blast furnace                      c) open hearth furnace  
b) Puddling furnace                  d) None of the above

- Q.6 The purest form of iron is:  
a) Cast iron                              c) Steel  
b) Wrought iron                        d) None of the above

- Q.7 Hematite is the ore of:  
a) Copper                                c) Iron  
b) Aluminium                          d) none of these

- Q.8 Steel is an alloy of:  
a) Iron and carbon                      c) carbon and hydrogen  
b) Iron only                                d) None of these

- Q.9 Brass is an alloy having.  
a) Copper and zinc                      c) both a & b  
b) Copper and Tin                        d) None of these

- Q.10 Ceramics belongs to category of:  
a) Inorganic non-metallic material  
b) Composite material  
c) Ferrous material  
d) All of the above

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name any one importance of engineering materials.

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- Q.12 Name any one cutting tool material.  
Q.13 Give any two example of non ferrous metals  
Q.14 Define non metallic material  
Q.15 Name any two glass types  
Q.16 LDPE stands for \_\_\_\_\_  
Q.17 Define malleability  
Q.18 PVC full form is \_\_\_\_\_  
Q.19 Define ceramics  
Q.20 Name thermal conductivity

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Classify engineering materials in brief.  
Q.22 Explain different types of cast iron and their properties in brief.  
Q.23 Classify steel? Also explain factor affecting its physical properties.  
Q.24 Explain classification of non-ferrous metals  
Q.25 Write the properties and uses of Nylon 66 polymer.  
Q.26 Write any two thermal and electrical properties of engineering materials.  
Q.27 Explain various specifications of engineering materials according to BIS  
Q.28 Write a short note on non metallic materials with example.

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